

Project Title	Telecom AI Brand Intelligence System
Skills Takeaway From This Project	Python, NLP, Deep Learning, HuggingFace Transformers, Transfer Learning, Vector DB, RAG, Synthetic Data Generation, Streamlit, API Integration, Data Preprocessing, Model Fine-Tuning, Sentiment Analysis, Prompt Engineering, LLM Integration, Data Visualization
Domain	Telecom Industry, Customer Experience Analytics, Brand Intelligence AI

Problem Statement:

ZENDS Communications is a virtual telecom company providing mobile, broadband, cloud, and IoT services. Telecom companies continuously receive large volumes of customer feedback from multiple public channels such as social media posts, mobile app reviews, online surveys, support platforms, and customer feedback portals.

These customer opinions are typically unstructured text and arrive in large volumes. Analyzing them manually is difficult, time-consuming, and inefficient. Without automated systems, telecom companies struggle to understand customer sentiment, detect recurring complaints, identify service issues early, and monitor overall brand perception.

To address this challenge, we develop a **Telecom AI Brand Intelligence System**, an AI powered platform that automatically analyzes customer feedback using Natural Language Processing and sentiment analysis.

The system classifies customer feedback into **positive, neutral, or negative sentiments** and identifies the **telecom service category** mentioned in the message.

The system also integrates **Retrieval-Augmented Generation (RAG)** to retrieve relevant telecom policies, troubleshooting guides, and company documents. Using this information, the AI system generates explanations and insights about potential reasons behind customer dissatisfaction.

The system contains **two interfaces designed for different users**.

User Interface

Customers can enter feedback or complaints through a **Streamlit-based interface**.

Once the user submits a message, the AI system automatically:

- Detects the sentiment (Positive / Neutral / Negative)
- Identifies the telecom service category related to the issue
- Retrieves relevant telecom policy or service documents using RAG
- Generates an AI response that understands the user's concern

The AI response is designed to acknowledge the customer concern politely and provide relevant information based on company policies.

Example response:

"We understand your concern regarding the service delay. Based on our service policy, activation requests may take up to 7 days to process. Our team is currently working to resolve the issue. If you need further assistance, please let us know."

This ensures that responses are **polite, respectful, and aligned with telecom policies**

Telecom Brand Analyst Dashboard

The second interface is designed for the **Telecom Brand Analysis Team**.

This dashboard provides real-time analytical insights from customer feedback, including:

- Distribution of positive, neutral, and negative sentiments
- Topic-wise complaint frequency
- Service category sentiment breakdown
- Real-time monitoring of new feedback

The system also generates **AI-based recommendations for improving telecom service quality** by analyzing patterns in customer complaints and referencing company policies through the RAG system.

These insights help telecom companies make **data-driven decisions to improve customer experience and brand perception**.

Business Use Cases:

- Monitor telecom brand perception using automated sentiment analysis.
- Identify recurring customer complaints related to telecom services.
- Detect service issues early through feedback trend analysis
- Improve customer experience by understanding customer concerns.
- Provide AI-generated insights for service quality improvement.
- Simulate enterprise telecom brand analytics using synthetic datasets.

Approach:

Step 1: Virtual Company & Dataset Design

- Created a virtual telecom company: ZENDS Communications.
- Defined telecom service categories such as Mobile Network, Broadband Service, Billing & Payments, Customer Support, Service Activation.
- Designed realistic telecom customer feedback messages that simulate real customer reviews and complaints.
- Rationale: In industry, structured datasets are rarely available initially. Synthetic data allows fast model development while maintaining realism.

Step 2: Synthetic Dataset Generation

In this step, students need to generate a **synthetic telecom customer feedback dataset** that will be used for training and evaluating NLP models in the AI Brand Intelligence System.

The dataset should contain around **10,000 or more telecom customer feedback records** including **positive, negative, and neutral feedback**.

Dataset Columns:

The dataset should include the following fields:

- **Feedback_Text** – Customer feedback related to telecom services
- **Service_Category** – Category of telecom service.
- **Sentiment** – Sentiment label (Positive, Negative, Neutral, Mixed).

Dataset Generation Approaches

- Students are encouraged to explore different approaches for generating synthetic telecom feedback data, such as **template-based feedback generation**, **sentiment-controlled text generation**, **LLM-based synthetic data generation using prompt engineering**, and **paraphrasing/data augmentation using HuggingFace models**.
- Example **code notebooks** are provided for **template-based feedback generation** and **LLM-based synthetic data generation through prompt engineering** to help students understand the implementation process.
- These examples are provided **only as reference implementations**, and students may adapt or extend them while experimenting with their own dataset generation strategies.

Step 3: Sentiment Analysis & Topic Classification Modelling:

In this step, students develop AI components to **analyse telecom customer feedback and identify both sentiment and service category**.

Sentiment Categories

- **Positive**
- **Neutral**
- **Negative**

Students are encouraged to **experiment with multiple sentiment analysis approaches** and compare their advantages and trade-offs.

Approach 1: Custom Deep Learning Models

- Students can train their own sentiment analysis models using the **synthetic dataset generated in Step 2**.
- Possible models include: **LSTM, GRU, etc.**
- This approach helps students understand **how custom NLP models are built and trained for business use cases**.

Approach 2: Pre-Trained Transformer Models

- Students can leverage pre-trained transformer models from HuggingFace and Frontier Models (Optional).

Example models (Open Source):

- DistilBERT
- BERT
- RoBERTa
- DeBERTa

Example sentiment models:

- `distilbert-base-uncased-finetuned-sst-2-english`
- `cardiffnlp/twitter-roberta-base-sentiment`
- `nlptown/bert-base-multilingual-uncased-sentiment`

Students can either:

- Use the models directly for inference.
- Fine-tune them using the synthetic telecom dataset

These models usually provide better contextual understanding and higher accuracy.

Approach 3: LLM-Based Sentiment Analysis

- Students can also perform sentiment analysis using Large Language Models (LLMs) with prompt engineering.
- Example prompt:

"Classify the following telecom customer feedback into Positive, Neutral, or Negative sentiment."

- This approach demonstrates sentiment analysis without building or training a traditional model.

Topic Classification

For topic classification, the system leverages LLM reasoning ability instead of training a separate model.

In this approach:

- The LLM analyzes the feedback text
- The LLM selects the most relevant category

Example prompt:

"Based on the customer feedback below, identify which telecom service category it belongs to from the following list: Mobile Network, Broadband Service, Billing & Payments, Customer Support Service Activation, General Complaint.

Return only the category name.

The LLM then returns only the category label.

Example output:

Billing & Payments"

This approach simplifies system design because:

- No additional topic classification model needs to be trained
- LLM reasoning identifies the correct category
- Categories can be updated without retraining models

Note: Students are not restricted to using an LLM for this task. They may also choose to build a separate topic classification model. In that case, they would need to generate or collect a labeled dataset, possibly using synthetic data generation, and then train a classification model. The choice of approach depends on the student's design decision and learning objective.

Model Evaluation: Students evaluate sentiment models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, and compare different approaches based on accuracy, computational cost, and ease of implementation.

Step 4: Retrieval-Augmented Generation (RAG)

- Convert company policy and service documents into embeddings (Sentence-Transformers).
- Store embeddings in vector database. Options (Fasis, Pinecone, Chroma)
- Retrieve relevant context based on customer query.
- Generate explanations and recommended responses using open-source HuggingFace LLMs (examples: Mistral-7B, Falcon-7B, LLaMA-2-7B) for free or lightweight inference. Can also explore paid LLM APIs based on the project cost.

Step 5: Streamlit Application

Develop a Streamlit interactive application.

Features include:

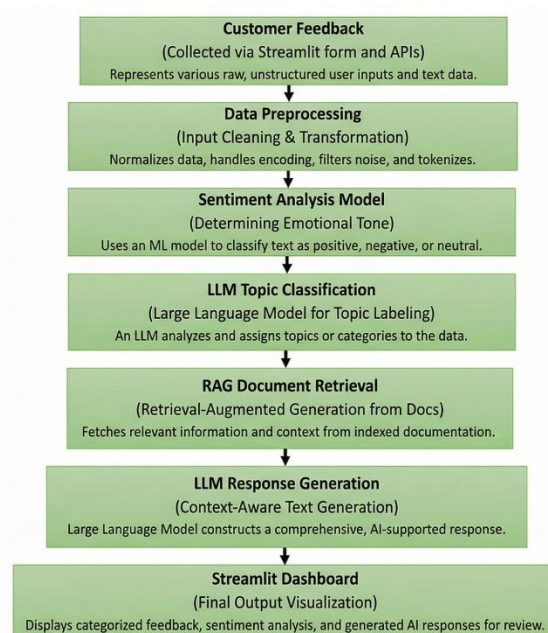
User Interface

- Accepts customer feedback input and detects service category and sentiment.
- Shows AI-generated explanation and response.

Brand Analyst Dashboard

- Displays sentiment distribution charts
- Shows topic-wise complaint frequency
- Provides real-time monitoring of feedback
- Shows AI-generated recommendations for service improvement

Data Flow and Architecture:



Exploratory Data Analysis (EDA)

- Univariate Analysis: Sentiment distribution, topic distribution.
- Bivariate Analysis: Sentiment vs service category.
- Multivariate Analysis: Message length vs sentiment vs topic.
- Outlier detection: unusually long/short messages, rare topic.
- Correlation: sentiment frequency per topic, high-severity complaints.
- Business Insights: Students should identify and present the most impactful insights suitable for a real-time dashboard, explain how these insights could affect business decisions, and justify why those insights were selected.

Results:

- Created a balanced synthetic telecom feedback dataset.
- Implemented multiple sentiment analysis approaches.
- Leveraged LLM for topic classification.
- Built a RAG-based explanation system.
- Developed an interactive Streamlit dashboard.
- Provided trade-off analysis for the chosen approaches.

Project Evaluation Metrics:

- Sentiment classification accuracy.
- Model F1-Score.
- Relevance of retrieved documents in RAG.
- Quality of AI-generated responses.
- Usability of Streamlit dashboard.

Technical Tags:

Python, NLP, Deep Learning, HuggingFace Transformers, Vector DB, RAG, Streamlit, Synthetic Data Generation, Prompt Engineering, LLM Integration

Deliverables:

- Synthetic telecom feedback dataset.
- Sentiment analysis models.
- LLM-based topic classification module.
- Vector database of telecom policy documents.
- RAG-based AI response generator.
- Streamlit dashboard.
- Complete project documentation.

Timeline:

Check your email for submission deadlines related to this project.

Project Learning Outcomes:

- Simulate real enterprise data by generating telecom customer feedback datasets based on company operations and service scenarios.
- Integrate NLP models for sentiment classification to automatically detect customer opinions from unstructured text data.

- Build a Retrieval-Augmented Generation (RAG) based AI system using vector databases and open-source Large Language Models (LLMs).
- Integrate deep learning models into a functional web application using Streamlit to create an interactive AI-powered platform.
- Learn industry best practices in synthetic dataset creation and enterprise AI system design by working with a domain-driven problem and company context.
- Develop the ability to solve real-world AI problems even without structured company datasets by generating and utilizing synthetic data effectively.

Problem Solved:

- Demonstrates the operations and challenges of a telecom company through a simulated enterprise environment.
- Automates customer feedback analysis and support responses, reducing the effort required to manually analyze large volumes of customer messages.
- Improves response efficiency by enabling AI to understand and classify customer complaints automatically.
- Provides policy-compliant, context-aware responses using the RAG system and telecom service documentation.
- Shows an end-to-end enterprise AI workflow, including data generation, NLP modeling, LLM integration, and dashboard deployment.
- Prepares students for industry-ready AI system development using technologies such as NLP, deep learning, prompt engineering, and RAG architectures.

Project Reference Documents:

All required documents and sample code file given in the final project Folder.